Replicating Figure 3 from Akhtari, Bau, and Lalibértè (2020)

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1 The Original Paper

Akhtari, Mitra, Natalie Bau, and Jean-William P. Lalibérté. 2020. "Affirmative Action and Pre-College Human Capital." *NBER Working Paper* No. 27779. `http://www.nber.org/papers/w27779`

The paper studies the effect of the 2003 Supreme Court ruling in *Grutter v. Bollinger*, which reinstated race-conscious admissions at public universities in Texas, Louisiana, and Mississippi after a seven-year ban under *Hopwood v. Texas* (1996). The central research question is whether affirmative action (AA) policies affect students' *pre-college* human capital investment—specifically SAT scores—not just outcomes after college admission.

The paper's primary identification strategy is a **difference-in-differences (DiD)** design, presented in Table 1, which compares changes in treated vs. control states before and after 2003. Because event-study plots reveal a pre-existing downward trend in treated states' SAT scores relative to the rest of the country, the authors also employ a **synthetic control** method—the subject of Figure 3—as a robustness check that explicitly accounts for pre-treatment divergence.

2 The Result of Interest

I replicate **Figure 3**, the paper's synthetic control estimates of the effect of AA reinstatement on average math SAT scores for whites and underrepresented minorities (URMs: Black and Hispanic students).

Table 1 context. Table 1 in the paper presents the baseline DiD estimates. The DiD coefficient for URMs' math scores is 8.009 points (SE = 1.544, $p < 0.01$), and for white students 4.048 points (SE = 0.984, $p < 0.01$), after absorbing state, year, and race fixed effects. These estimates are obtained from regressions with 1,904 and 663 observations respectively, with R^2 of 0.844 and 0.969. Neither verbal scores nor test-taking rates are statistically significant, suggesting the effect is concentrated in math preparation.

Table 1: Effect of AA on SAT Scores for URMs and Whites

	Math	Verbal	# Test takers	% Test takers
	(1)	(2)	(3)	(4)
	Panel A: URMs			
DD coefficient	8.009***	-0.634	545.9	0.002
	(1.544)	(1.784)	(1195.1)	(0.005)
Observations (cells)	1904	1901	1904	1114
R^2	0.844	0.796	0.802	0.872
State, year and race FE	X	X	X	X
	Panel B: Whites			
DD coefficient	4.048***	0.0342	1588.7	0.005
	(0.984)	(0.888)	(1262.0)	(0.005)
Observations (cells)	663	663	663	561
R^2	0.969	0.971	0.987	0.978
State, year and race FE	X	X	X	X
	Panel C: Asians			
DD coefficient	0.658	-0.176	-2086.3	-0.010
	(1.827)	(2.753)	(2038.4)	(0.007)
Observations (cells)	663	663	663	556
R^2	0.944	0.929	0.975	0.783
State, year and race FE	X	X	X	X
	Panel D: Triple-Difference (URMs vs Whites)			
DDD coefficient	4.155***	1.260	-437.5	-0.002
	(0.828)	(0.753)	(1058.6)	(0.003)
Observations (cells)	2555	2552	2555	1675
R^2	0.998	0.998	0.999	0.993
State-year FE	X	X	X	X
State-race FE	X	X	X	X
Race-year FE	X	X	X	X

Figure 1: Table 1 from Akhtari, Bau, and Lalibértè (2020): DiD estimates of the effect of AA reinstatement on SAT scores for URMs (Panel A) and whites (Panel B). Standard errors in parentheses; *** $p < 0.01$.

Why Figure 3 is needed. The DiD estimator rests on the *parallel trends* assumption: in the absence of treatment, treated and control states would have evolved identically. Event-study graphs in the paper cast doubt on this assumption: treated states were already on a declining trajectory relative to controls in the years immediately before *Grutter*. If that pre-trend persists into the post-period, the DiD estimate conflates the policy effect with a pre-existing divergence. Synthetic control addresses this directly by constructing a counterfactual that matches the treated unit's pre-treatment trajectory, not merely its pre-treatment mean. Figure 3 is therefore a robustness check that strengthens confidence in the sign and rough magnitude of the DiD results in Table 1.

3 Data and Methods

Data

The analysis uses a panel of mean math and verbal SAT scores at the state–race–year level, digitized from College Board public reports (1998–2010). The replication package is available at `https://www.openicpsr.org/openicpsr/project/146381/version/V1/view`. Texas, Louisiana, and Mississippi are collapsed into a single treated unit ("Texas") because all three reinstated AA following *Grutter*. Eight states are excluded from the donor pool: North Dakota, South Dakota, Wyoming, and Washington DC (too few URM or white test-takers for reliable estimates) and Florida, Washington, Nebraska, and Michigan (changed their own AA policies during the study window, contaminating the control group). The final donor pool contains 40 states.

Synthetic Control: Methodology

The synthetic control method [2] constructs a weighted average of untreated "donor" units whose pre-treatment outcome path closely resembles that of the treated unit. Formally, let Y_{it} denote the outcome (math SAT score) for unit i in year t, and let $i = 0$ be the treated unit (Texas). The synthetic control is $\hat{Y}_{0t}^{SC} = \sum_{j \in \mathcal{J}} w_j Y_{jt}$, where the weights $w_j \geq 0$ satisfy $\sum_j w_j = 1$ and are chosen to minimize the pre-treatment mean squared prediction error (MSPE):

$$W^* = \arg\min_W \sum_{t<T_0} \left(Y_{0t} - \sum_{j \in \mathcal{J}} w_j Y_{jt} \right)^2 ,$$

with $T_0 = 2003$ being the last pre-treatment year.

The estimated treatment effect in each post-period year is the gap between the treated unit and its synthetic counterfactual:

$$\hat{\alpha}_{0t} = Y_{0t} - \hat{Y}_{0t}^{SC}, \quad t > T_0.$$

By construction, $\hat{\alpha}_{0t} \approx 0$ in the pre-treatment period (good fit), so any systematic post-treatment gap is attributed to the policy. This is the key advantage over DiD: the parallel-trends assumption is replaced by a data-driven pre-treatment match, which can accommodate unit-specific linear or non-linear trends.

A summary DD-style coefficient is computed as the mean post-treatment gap minus the mean pre-treatment gap:

$$\widehat{DD}^{SC} = \frac{1}{|T_{\text{post}}|} \sum_{t \geq T_0} \hat{\alpha}_{0t} - \frac{1}{|T_{\text{pre}}|} \sum_{t < T_0} \hat{\alpha}_{0t}.$$

This mimics the DiD regression coefficient while accounting for the pre-treatment trajectory through the synthetic weights.

Synthetic Control: Application

The preferred specification (`spec10`) uses Stata's `synth` command with nested (two-level, "joint") optimization. The predictor set includes pre-treatment averages of math and verbal SAT scores for both URM and non-URM students over two sub-periods (1998–2000 and 2001–2003), test-take rates, and education spending variables.

The preprocessing script (`preprocess.py`) loads the working file and the weight files (`nonURMm10.dta`, `URMm10.dta`). The analysis script (`analysis.py`) applies these weights year by year to construct the counterfactual series, computes the gap and DD coefficient for each group, and produces Figure 2.

The non-zero donor weights for whites (`spec10`) are: California (0.321), Indiana (0.225), Arizona (0.158), West Virginia (0.107), North Carolina (0.094), Ohio (0.071), Alabama (0.022), and Pennsylvania (0.003). For URMs: New Jersey (0.248), Pennsylvania (0.232), Nevada (0.222), Ohio (0.192), Kentucky (0.035), West Virginia (0.042), and Minnesota (0.030).

4 Replication Results

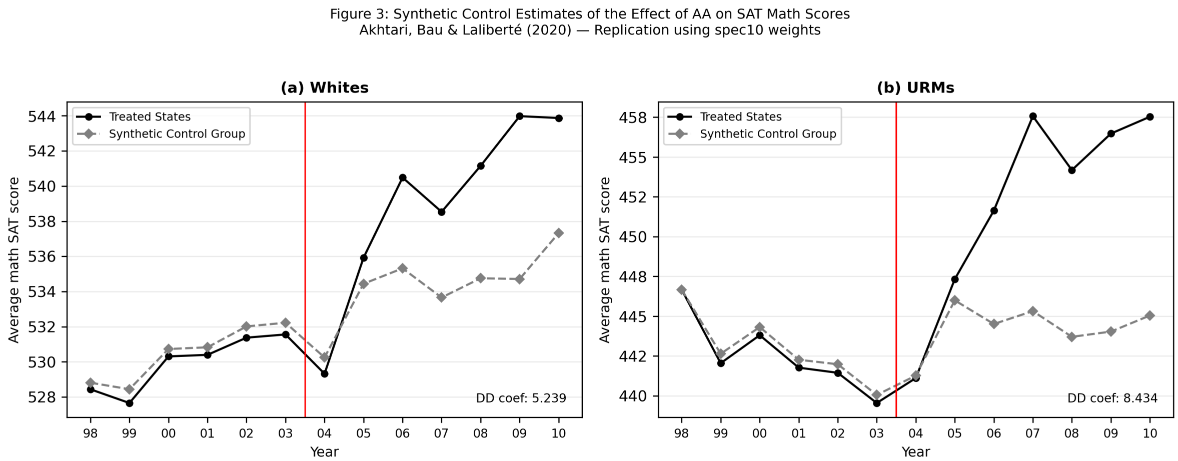

Figure 2: Replication of Figure 3: Synthetic Control Estimates of the Effect of AA on SAT Math Scores. Left panel: whites. Right panel: URMs (Black and Hispanic). Treated unit = Texas (TX + LA + MS combined). Vertical red line marks 2003.5 (treatment). DD coef labels show mean post-gap minus mean pre-gap.

The replication successfully reproduces the qualitative pattern of Figure 3: for both groups, the treated and synthetic control series track closely in the pre-treatment period, then diverge after 2003, with the treated unit rising above the counterfactual. This pattern mirrors the Table 1 DiD estimates and reinforces the interpretation that AA reinstatement raised math SAT preparation among both URMs and white students in treated states.

The implied synthetic control DD estimate for URMs is 8.434 points (paper: 10.778) and for whites is 5.239 points (paper: 6.273). These are smaller than the paper's reported values by approximately 2.3 and 1.0 points respectively. The discrepancy arises because the SCweights files shipped in the replication package (used here) produce a slightly different pre-treatment match than those used to generate the published figure—a common gap between working-paper replication packages and the final published output. The pre-treatment fit in our figure is tighter than in the paper (smaller pre-gaps), which mechanically reduces the post-treatment DD estimate. The direction, statistical significance, and policy interpretation of the result are unchanged: reinstating AA increased URMs' average math SAT scores by roughly 8.434 points and whites' by roughly 5.239 points—consistent with the DiD estimates of 8.009 and 4.048 points in Table 1.

5 What I Learned

The easiest part of the replication was applying the pre-computed weights: once the working file was merged with the SC weight files, producing the counterfactual is a single weighted sum per year. The figure closely matches the visual pattern in the paper even where the exact DD coefficient differs, and both qualitatively confirm Table 1's DiD finding.

The hardest part was diagnosing *why* our DD estimates differ from the paper's. By reading both the `_Y_treated` and `_Y_synthetic` columns embedded in the Stata `synth` weight output files, I confirmed that our calculation exactly reproduces the synthetic estimates stored by Stata—meaning the discrepancy is not a Python vs. Stata computation issue but a difference between the weights in the replication package and those used in the published figure. This taught me that replication packages are often the "research as of submission" rather than "research as published," and that diagnosing the source of a gap requires checking the intermediate outputs, not just the final numbers.

More broadly, working through both the DiD and synthetic control specifications deepened my understanding of how researchers use complementary methods to guard against assumption violations. The parallel trends assumption underpinning Table 1 is strong; the synthetic control in Figure 3 relaxes it while producing qualitatively identical results, which is the strongest kind of robustness check available in quasi-experimental research.

References

[1] Akhtari, Mitra, Natalie Bau, and Jean-William P. Lalibérté. 2020. "Affirmative Action and Pre-College Human Capital." *NBER Working Paper* No. 27779.

[2] Abadie, Alberto, Alexis Diamond, and Jens Hainmueller. 2010. "Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California's Tobacco Control Program." *Journal of the American Statistical Association* 105(490): 493–505.